MILITARY SPECIFICATION

CHAIR, HOSPITAL WARD, METAL

1. SCOPE

1.1 This specification covers one type of armless chair, of tubular steel construction, suitable for hospital ward use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-491	Enamel; Gloss, Synthetic (for Metal and Wood Hospital Furniture.
TT-P-659	Primer-Surfacer; Synthetic, Tints and White, (for Metal and Wood Surfaces).
CCC-A-700	Artificial Leather, Cloth, Coated, Vinyl Resin, (Upholstery).
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple, Wall.

Military

MIL-R-5001	Rubber; Latex Foam Sponge.
MIL-R-17252	Rubber, Cellular (Bonded Shredded).

FSC 6530

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STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods.
Fed. Std. No. 595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Medical Supply Center

1834 Chair, Hospital Ward, Metal

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein and shall be tested as specified in 4.4. Material not definitely specified shall be of the best quality used for the purpose in commercial practice. All materials used in the construction of the chair shall be new. Gages specified herein are minimum. Commercial tolerances shall apply.

3.1.1 Tubing. The leg bows and back uprights shall be constructed of 1-1/8-inch square tubing having a wall thickness of 0.035 inch (No. 20, Birmingham Wire (B.W.) gage). The stretchers shall be of 1- by 1/2-inch rectangular tubing having a wall thickness of 0.035 inch (No. 20, B.W. gage). The tubing shall be made of cold-rolled strip steel, rolled to shape, the edges butt-jointed and electric-resistance welded. The welded seam shall be sound and smooth. The corners of all tubing shall be well rounded. Not lower than American Iron and Steel Institute (AISI) 1010 steel shall be used for the square tubing and the rectangular tubing. A tolerance of $\pm$ 1/32 inch in outside dimensions of tubing shall be acceptable.

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3.1.2 Sheet. Steel sheet used in fabricating the seat pan and back form shall be AISI 1010 steel. The metal used shall be 0.0478 inch thick (No. 18, United States Standard gage).

3.1.3 Artificial leather (upholstery). The artificial leather for upholstery purposes shall be synthetic resin coated conforming to class 2 of Specification CCC-A-700, except that accelerated weathering volatility and plasticizer loss tests shall not be required. The material shall be dark green (see 6.2).

3.1.4 Rubber.

3.1.4.1 Seat. Rubber for the seat shall be as specified in 3.1.4.1.1 or 3.1.4.1.2 at the option of the manufacturer.

3.1.4.1.1 Foam rubber. Foam rubber shall conform to Specification MIL-R-5001 for type II, class medium, grade C, 1-1/4 inches thick.

3.1.4.1.2 Bonded shredded foam rubber. Bonded shredded foam rubber shall conform to Specification MIL-R-17252 for type II, class 2, medium condition, 1-1/4 inches thick, except that material shall be made entirely of foam latex rubber particles capable of passing a 3/16-inch mesh wire screen. In addition, the top surface and edges of the seat slab shall be given an even coating of foam rubber cement which shall give the slab a skinlike finish and shall not penetrate below the applied surfaces (see 6.3).

3.1.4.2 Back. Foam rubber shall conform to Specification MIL-R-5001 for type II, class soft, grade C, 1 inch thick.

3.1.5 Cotton felt. The cotton felt shall be made of unbroken laminated sheets of garnetted felt made from a blend of not less than 25 percent cotton not lower in grade than any of the following United States Standard grades: No. 7, or Low Middling; No. 7, or Low Middling Spotted; or No. 4, or Strict Middling Grade, having a staple length of not less than 1/2 inch; and 75 percent linters not lower in grade than United States Standard No. 3 Middling. Thickness of the felt shall be not less than 3/8 inch.

3.1.6 Glides. Glides shall be of all metal construction with a rubber cushion. Glides shall fit snugly in the legs. Glide bases or domes shall be AISI 302 corrosion-resisting steel. Other steel parts shall be zinc or cadmium plated.

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3.1.7 Fastening devices. Fastening devices, such as screws and tee nuts, shall conform to standard commercial requirements and shall be zinc or cadmium plated.

3.1.8 Primer. Synthetic primer shall conform to Specification TT-P-659.

3.1.9 Enamel. Enamel shall conform to class B of Specification TT-E-491, except that phthalocyanine pigments may be used to obtain the specified color. The color shall be light green, conforming to color 14241 of Standard No. 595 (see 6.1).

3.2 Construction. The chair shall consist of a chair frame and removable, padded, and upholstered steel seat and back sections. All details of design and construction shall be as specified herein and as shown in Drawing 1834.

3.2.1 Chair frame. The chair frame shall consist of two leg bows, two uprights, and three stretcher tubes joined by continuous bronze welding to form a strong, rigid assembly. All corners shall be well rounded. Edges, welds, and joints shall be free of scratching and tearing hazards. The welds shall be finished off smoothly to present a proper surface for finishing.

3.2.1.1 Leg bows. The leg bows shall each be made from a continuous length of 1-1/8-inch square steel tubing (see 3.1.1) shaped as shown on the drawing.

3.2.1.2 Stretchers. The leg bows shall be joined and reinforced by means of three rectangular steel stretcher tubes (see 3.1.1), located at the front and back of the seat between the rear legs in the positions shown on the drawing.

3.2.1.3 Uprights. Each upright shall be constructed of a continuous length of 1-1/8-inch square steel tubing (see 3.1.1), shaped as shown on the drawing. The lower end of each upright shall be continuously bronze welded to the rear seat stretcher tube, and each upright shall be bronze welded at the top and bottom to the rear leg section of the appropriate leg bow at the point of maximum curvature. Length of welds, where the upright is joined to the rear leg, shall be not less than 1-1/4 inches.

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3.2.2 Upholstering forms.

3.2.2.1 Seat. The seat pan shall be stamped from steel sheet (see 3.1.2) and shaped and punched as shown in the drawing. All edges of the pan shall be hemmed to eliminate sharp edges. The finished seat pan shall be padded with rubber (see 3.1.4.1), evenly and securely cemented thereto, followed by a layer of cotton felt (see 3.1.5), and then upholstered with a final covering of artificial leather (see 3.1.3). The final covering shall be tailored with a welted box seam, continuous around two sides and the back, and shall cover the seat smoothly and be held in position by cement and integral lugs formed by perforating the seat pan. There shall be three lugs at each corner of the seat pan as shown in the drawing, and each side of the pan shall have, in addition, a minimum of six additional lugs arranged for maximum security of the final covering. The central portion of the seat pan shall be provided with four round perforations to permit venting of the upholstery. The upholstered form shall be a complete assembly for attachment to the frame, and shall be attached by four 1/4-inch machine screws extending through the leg bow tubing into tee nuts welded to the inside surface of the seat pan.

3.2.2.2 Back. The back upholstering form shall be fabricated of sheet steel (see 3.1.2) and shall be shaped and punched as shown in the drawing. The back form shall be upholstered in a similar manner to the seat, except that rubber shall be as specified in 3.1.4.2 and upholstery fabric shall smoothly cover all surfaces (front and back) of the back upholstering form. In addition to the three integral lugs at each corner of the form, there shall be a minimum of four additional lugs for each of the shorter sides, and a minimum of seven additional lugs for each of the longer sides of the back form to hold the upholstering artificial leather in place. The upholstery covering shall be made of four pieces. The two side pieces shall be joined to the front piece with welted box seams and the back piece shall be joined to the top of the front piece with a plain inside box seam. The remaining three sides of the back piece shall be hemmed and stitched. All leather in contact with the metal surfaces of the back form shall be securely cemented in place. A cement suitable for securely adhering vinyl to metal shall be used (see 6.4). A 1/8-inch diameter rod shall be inserted through the bottom hem of the back cover and shall be of such length that it projects under and is secured by the uprights. The rod shall be zinc or cadmium plated. The upholstered form shall be a complete assembly for attachment to the frame uprights, and shall be attached to the uprights by means of two 1/4-inch machine screws extending through the upright tubing into tee nuts securely welded to the inside surface of the back form.

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3.3 Finish.

3.3.1 Cleaning. Cleaning shall be accomplished as specified for type I of Specification TT-C-490. All welds shall be thoroughly and completely cleaned of flux, scales, etc., prior to application of primer.

3.3.2 Primer. After cleaning, the parts shall be given one coat of synthetic primer (see 3.1.8). The primer shall be baked on for not less than 45 minutes at a temperature of 250° Fahrenheit (F.), or for an equivalent baking period at a higher or lower temperature, as recommended by the paint supplier and approved by the contracting officer. Variations in temperature and time of baking approved for a previous contract are not required to be submitted for approval. The primer shall be applied so as to level out and produce a uniform surface free of pebbles and wrinkles.

3.3.3 Enameling. After the primer has been allowed to cool, all surfaces except the back and seat pans shall then be covered with two single coats of enamel (see 3.1.9). Each coat shall be baked on separately for not less than 45 minutes at a temperature of 250° F., or for an equivalent baking period at a higher or lower temperature, as recommended by the paint supplier and approved by the contracting officer. Variations in temperature and time of baking approved for a previous contract are not required to be submitted for approval. The enamel coat shall be applied so as to level out and produce a final uniform surface without runs, wrinkles, orangepeel, or grit, and without areas of thin or no film. All surfaces of the back and the seat pans shall be given one coat of enamel (see 3.1.9), baked on in the same manner.

3.4 Identification marking. The manufacturer's name or registered trademark, the contract or purchase order number, the date of manufacture, and the letters U.S., in 1/4-inch characters, shall be stamped or stenciled with black, indelible, waterproof ink on the underneath surface of the seat pan. All characters shall be legible. Marking secured by adhesives shall not be used.

3.5 Workmanship. Workmanship shall be first class throughout. Each chair shall be rigid in assembly, protected throughout against corrosion, and clean, well finished, and free from any defects which affect serviceability or appearance. All edges of the metal members shall be straight, clean, and free of burrs. Chairs shall be free of openings which may act as vermin traps. Work shall be accomplished by experienced operators, adequately supervised and carefully inspected.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105.

Component	Characteristic	Requirement	Test procedure
Artificial leather	Physical properties	3.1.3	4.4.2
Rubber	Physical properties	3.1.4	4.4.2
Enamel *	Physical properties	3.1.9	4.4.2

* Sufficient test panels prepared in accordance with the procedure outlined in 3.3 shall be submitted for tests outlined in 4.4.2.

4.2.3 Raw materials. Four ounces of each lot of each metal used in the construction of the chairs shall be selected for chemical analysis.

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4.3 Examination. The chairs shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Federal Test Method Standard No. 151.

4.4.2 Components. Tests of the component materials used in the manufacture of the chairs shall be in accordance with the quality assurance provisions of their respective specifications, as listed below:

Artificial leather CCC-A-700
Rubber (compression resistance, density, recovery, and accelerated aging only)
  Foam MIL-R-5001
  Bonded shredded foam MIL-R-17252
Enamel .....TT-E-491 (except F-2a)

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level A.

5.1.1 Unit of issue. One (1) chair, as specified, constitutes one unit of issue.

5.1.2 Unit preparation. All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.1.2.1 Arrangement. Two chairs shall be packed per container. One chair shall be inverted with the seats of the two chairs together, the backs on opposite sides, and the legs at opposite ends of the container. The nested chairs shall be tied together with twine or other suitable materials.

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5.2 Packing.

5.2.1 Level A. Two chairs (2 units), arranged as above, shall be overpacked in an exterior container designed and constructed in accordance with Specification PPP-B-640, class 2, grade A, style E. The chairs shall be adequately secured within the exterior container by means of wood braces, blocks, or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

NOTE. Closure shall be in accordance with 30.3.5, except that the use of steel strapping is not authorized. Sealing shall be in accordance with 30.2 by the use of tape as specified in 30.1.5. Tape shall not be required on manufacturer's joint.

5.3 Marking. Exterior container shall be marked in accordance with Standard MIL-STD-129. In addition, the letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated.

6. NOTES

6.1 Availability of color chips. The individual 3- by 5-inch chips of the color referenced in 3.1.9 are available from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W., Washington 25, D.C.

6.2 The artificial leather referenced in 3.1.3 may be obtained from Du Pont as their Fabrilite V-6028, color 478EH, grain S-1, Spanish finish.

6.3 The rubber cement referenced in 3.1.4.1.2 may be obtained from Firestone, their Fomex rubber cement.

6.4 The cement referenced in 3.2.2.2 is available from Hadley Brothers, UHL, St. Louis, Missouri, as their #3642A.

6.5 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.6 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - BuMed
Air Force. - AFMSHB(03)

Review activities:
Army. - MD
Navy. - BuMed
Air Force. AFMSHB(03)

Preparing activity:
Defense Supply Agency. - DM

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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: CHAIR, HOSPITAL WARD, METAL--- MIL-C-36250(DM) dated 6 August 1964

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

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